YASKAWA

YRC1000micro OPTIONS
INSTRUCTIONS

FOR SERVOFLOAT FUNCTION

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

MOTOMAN INSTRUCTIONS

MOTOMAN-□□□ INSTRUCTIONS
YRC1000micro INSTRUCTIONS
YRC1000micro OPERATOR'S MANUAL
YRC1000micro MAINTENANCE MANUAL
YRC1000micro ALARM CODES (MAJOR ALARMS) (MINOR ALARMS)

Please have the following information available when contacting Yaskawa Customer Support:
- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
  Robot Serial Number *(Located on robot data plate)*
  Robot Sales Order Number *(Located on controller data plate)*

Part Number: 181294-1CD
Revision: 1

Yaskawa Customer Support
- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
HW1484512 ①

DANGER

- This manual explains the servofloat function of the YRC1000micro system. Read this manual carefully and be sure to understand its contents before handling the YRC1000micro. Any matter not described in this manual must be regarded as "prohibited" or "improper".
- General information related to safety are described in "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS. To ensure correct and safe operation, carefully read "Chapter 1. Safety" of the YRC1000micro INSTRUCTIONS.

CAUTION

- In some drawings in this manual, protective covers or shields are removed to show details. Make sure that all the covers or shields are installed in place before operating this product.
- YASKAWA is not responsible for incidents arising from unauthorized modification of its products. Unauthorized modification voids the product warranty.

NOTICE

- The drawings and photos in this manual are representative examples and differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice when necessary due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- If your copy of the manual is damaged or lost, contact a YASKAWA representative to order a new copy. The representatives are listed on the back cover. Be sure to tell the representative the manual number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installation, operation, maintenance, or inspection of the YRC1000micro.

In this manual, the Notes for Safe Operation are classified as “DANGER”, “WARNING”, “CAUTION”, or “NOTICE”.

DANGER

Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. Safety Signs identified by the signal word DANGER should be used sparingly and only for those situations presenting the most serious hazards.

WARNING

Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury. Hazards identified by the signal word WARNING present a lesser degree of risk of injury or death than those identified by the signal word DANGER.

CAUTION

Indicates a hazardous situation, which if not avoided, could result in minor or moderate injury. It may also be used without the safety alert symbol as an alternative to “NOTICE”.

NOTICE

NOTICE is the preferred signal word to address practices not related to personal injury. The safety alert symbol should not be used with this signal word. As an alternative to “NOTICE”, the word “CAUTION” without the safety alert symbol may be used to indicate a message not related to personal injury.

Even items described as “CAUTION” may result in a serious accident in some situations.
At any rate, be sure to follow these important items.

NOTE

To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” and “CAUTION".

DANGER

- Before operating the manipulator, make sure the servo power is turned OFF by performing the following operations. When the servo power is turned OFF, the SERVO ON LED on the programming pendant is turned OFF.
  - Press the emergency stop button on the programming pendant or on the external control device, etc.
  - Disconnect the safety plug of the safety fence. (when in the play mode or in the remote mode)

If operation of the manipulator cannot be stopped in an emergency, personal injury and/or equipment damage may result.

Fig. : Emergency Stop Button

- Before releasing the emergency stop, make sure to remove the obstacle or error caused the emergency stop, if any, and then turn the servo power ON.

Failure to observe this instruction may cause unintended movement of the manipulator, which may result in personal injury.

Fig. : Release of Emergency Stop

- Observe the following precautions when performing a teaching operation within the manipulator's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the manipulator’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the manipulator, which may result in personal injury.

- Confirm that no person is present in the manipulator's operating range and that the operator is in a safe location before:
  - Turning ON the YRC1000micro power
  - Moving the manipulator by using the programming pendant
  - Running the system in the check mode
  - Performing automatic operations

Personal injury may result if a person enters the manipulator's operating range during operation. Immediately press an emergency stop button whenever there is a problem. The emergency stop button is located on the right of the programming pendant.

- Read and understand the Explanation of the Warning Labels before operating the manipulator.

DANGER

- In the case of not using the programming pendant, be sure to supply the emergency stop button on the equipment. Then before operating the manipulator, check to be sure that the servo power is turned OFF by pressing the emergency stop button.
  Connect the external emergency stop button to the 4-14 pin and 5-15 pin of the Safety connector (Safety).
- Upon shipment of the YRC1000micro, this signal is connected by a jumper cable in the dummy connector. To use the signal, make sure to supply a new connector, and then input it.

If the signal is input with the jumper cable connected, it does not function, which may result in personal injury or equipment damage.

WARNING

- Perform the following inspection procedures prior to conducting manipulator teaching. If there is any problem, immediately take necessary steps to solve it, such as maintenance and repair.
  - Check for a problem in manipulator movement.
  - Check for damage to insulation and sheathing of external wires.
- Return the programming pendant to a safe place after use.

If the programming pendant is left unattended on the manipulator, on a fixture, or on the floor, etc., the Enable Switch may be activated due to surface irregularities of where it is left, and the servo power may be turned ON. In addition, in case the operation of the manipulator starts, the manipulator or the tool may hit the programming pendant left unattended, which may result in personal injury and/or equipment damage.

Definition of Terms Used Often in This Manual

The MOTOMAN is the YASKAWA industrial robot product.

The MOTOMAN usually consists of the manipulator, the YRC1000micro controller, manipulator cables, the YRC1000micro programming pendant (optional), and the YRC1000micro programming pendant safety signal short circuit connector (optional).

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
YRC1000micro controller	YRC1000micro
YRC1000micro programming pendant	Programming pendant (optional)
Cable between the manipulator and the controller	Manipulator cable
YRC1000micro programming pendant safety signal short circuit connector	Programming pendant safety signal short circuit connector (optional)

Descriptions of the programming pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. e.g. [ENTER]
	Axis Keys /Numeric Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, e.g. [SHIFT]+[COORD].
	Mode Switch	Mode Switch can select three kinds of modes that are denoted as follows: REMOTE, PLAY or TEACH. (The switch names are denoted as symbols)
	Button	The three buttons on the upper side of the programming pendant are denoted as follows: START, HOLD, or EMERGENCY STOP. (The button names are denoted as symbols)
	Displays	The menu displayed in the programming pendant is denoted with { }. e.g. {JOB}

Description of the Operation Procedure

In the explanation of the operation procedure, the expression "Select • • • " means that the cursor is moved to the object item and [SELECT] is pressed, or that the item is directly selected by touching the screen.

Registered Trademark

In this manual, names of companies, corporations, or products are trademarks, registered trademarks, or brand names for each company or corporation. The indications of (R) and TM are omitted.

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1 Servofloat Function

The servofloat function of the YRC1000micro consists of the link servofloat function and the linear servofloat function. The link servofloat function controls the torque independently for each axis; and the linear servofloat function controls the torque on the Cartesian coordinates. Select these two functions by setting the servofloat instructions accordingly.

1.1 Link Servofloat Function

The link servofloat function works in combination with the position control and the torque control, whereas an ordinary control is a position control that generates torque to keep the taught position regardless of the applied external force. “Torque control” keeps the taught position only by using the torque against the gravitational and frictional forces. Therefore, a manipulator moved by an external force will not return to its original position.

With the servofloat function, external force moves the manipulator since the torque control has priority over the position control.

For example, when removing a workpiece from the die-casting machine, a large external force is applied to the manipulator by the extruding cylinder when a workpiece is grasped and pulled out. The servofloat function is used to protect the manipulator from excessive external force, and the manipulator moves according to the force applied from the die-casting machine.

Note that the manipulator does not move exactly to the taught position or along the taught path, because complete position control is not performed during execution of the servofloat function.

1.2 Linear Servofloat Function

When a manipulator pushes a workpiece or visa-versa, a movement to one predetermined direction may be desirable. With the link servofloat function, a servofloat motion in one direction is not possible since torque control is performed independently for each axis.

This function can specify the control of force in one direction within a space.[1)]When unloading a die-casting machine, this function is useful for considerably long ejection distances or when polishing or grinding.

1.3 Using the Link Servofloat Function and the Linear Servofloat Function

Use the link servofloat function when high-ferquency noise, motor vibration, or unsteady manipulator motion occur depending on the manipulator posture, speed, and movement direction.

NOTE

Tool load information setting

- The servofloat function compensates for gravity in real-time in order to prevent the arms from dropping due to gravity even when the generated torque is limited. For this purpose, the tool load information used for gravity compensation must be set correctly. Otherwise, the servo-float function cannot be performed properly and the arms may drop down or rise up.
- For the tool load information, refer to “Chap.8.4.3 Tool Load Information Setting” in “YRC1000micro INSTRUCTIONS (RE-CTO-A222)” to correctly set the tool load information.

1 When external force is applied, the R, B, and T-axes move the same as in the link servo-float function.
Only the end of the U-axis moves in a linear motion.

2 The Servofloat Function

Enable the servofloat function with the servofloat function start instruction FLOATON. It remains enabled until one of the following conditions is established.

- Performing the a servofloat function end instruction FLOATOF
- Performing the FWD/BWD key operation, jogging, or a job startup after the cursor is moved
- Performing the FWD/BWD key operation, jogging, or a job startup after a job call, job selection, or job editing
- Turning the servo ON again

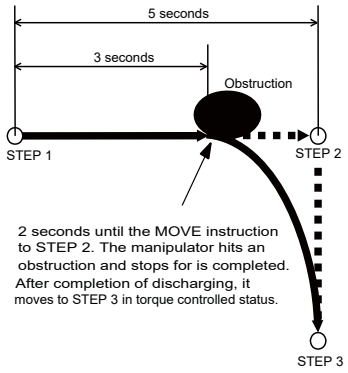
When the function is enabled, the manipulator generates only the torque that is less that the value set to the servofloat condition file. The manipulator uses the torque to operate and keep the position. If a large external force is applied, the manipulator moves according to this force.

2.1 Moving with the Move Instruction

If an external force prevents manipulator operation, the manipulator does not reach the taught position when the servofloat function is enabled. Even though the manipulator does not actually reach the taught position, the instruction that would be activated at that position are nonetheless enabled, so the move instruction terminated.

Therefore, when the manipulator cannot reach the taught position because of an external force, the manipulator performs the next instruction.

The following figure shows an example where the operation time is 5 seconds from STEP 1 to STEP 2. If the manipulator hits an obstruction 3 seconds after STEP 1 and stops there, it remains stopped at the position for the remaining 2 seconds, and then moves to STEP 3 without moving to STEP 2 when moving with the torque control.

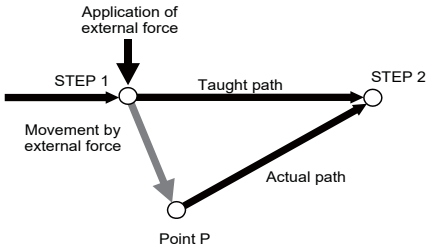



```
STEP 1   MOVL  V=100.0
         FLOATON  FL#(1)
STEP 2   MOVL  V=100.0
STEP 3   MOVL  V=100.0
         FLOATOF
```

2.2 Moving with the TIMER and WAIT Instructions

When the servofloat function is enabled, torque control can be performed even during the TIMER and WAIT instructions. These instructions end when time is up or when a condition described in page 2-1 is established.

For example, in the following figure, the manipulator reaches STEP 1, and the FLOATON instruction enables the servofloat function. Then the manipulator waits for the input of IN#1 by the WAIT instruction. In this state, if an external force is applied to the manipulator, the manipulator moves to the point P and stops. If the FLOATOF instruction is performed at this point, the servofloat function is disabled and the manipulator moves from point P to STEP 2.




```
STEP 1   MOVL  V=100.0
         FLOATON  FL#(1)
         WAIT  IN#(1)=ON
         FLOATOF
STEP 2   MOVL  V=100.0
```

2.3 Operation Check

When the servofloat function is enabled, the soft limit, cube interference, and the S-axis interference are checked at the actual position of the manipulator even though external force moved the manipulator.

2.4 When an Alarm Occurs

If an alarm occurs when the servofloat function is enabled, the servofloat function may remain effective depending on the type of alarm.

To disable the servofloat during an alarm, enter “external emergency stop” or “servo OFF signal” during the “alarm occurrence” signal output. This turns OFF the servo power supply and applies the brakes.

2.5 Display during Servofloat Operation

When the servofloat function is enabled, the message “Servo float activated” appears at the bottom of the programming pendant display, and the specific output #50510 to #50511 “SERVOFLOAT ON R1 to R2” is output.

2.6 Turning Servo ON Again During Servofloat Operation

When the servo power has been turned OFF due to an emergency stop during servofloat operation and turns ON again, press the [SERVO ON READY] on the playback panel until this servo recognizes that the power supply is turned ON. If the button is not pressed for a long enough time, the message “depress servo power ON” appears and the servo power cannot be turned ON.

3 Servofloat Condition Files

CAUTION

- If the values for the maximum force and maximum torque are set too low, gravity may cause an arm to drop.

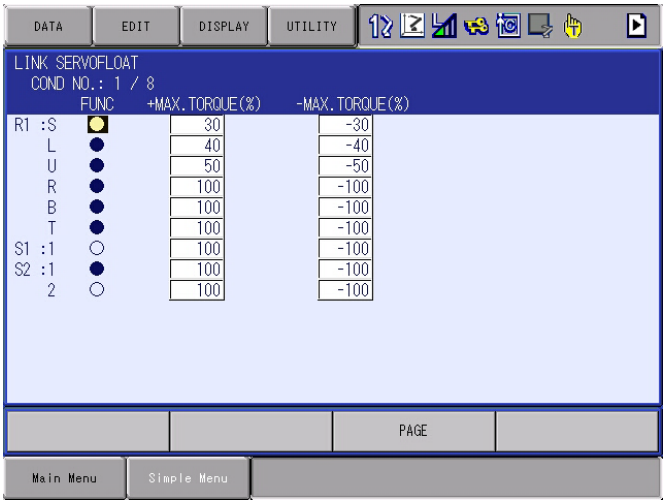
The arm drop may cause an accident or damage to the devices.

There are two kinds of servofloat condition files: the link servofloat condition file specified for the link servofloat function, and the linear servofloat condition file specified for the linear servofloat function.

3.1 Link Servofloat Condition File

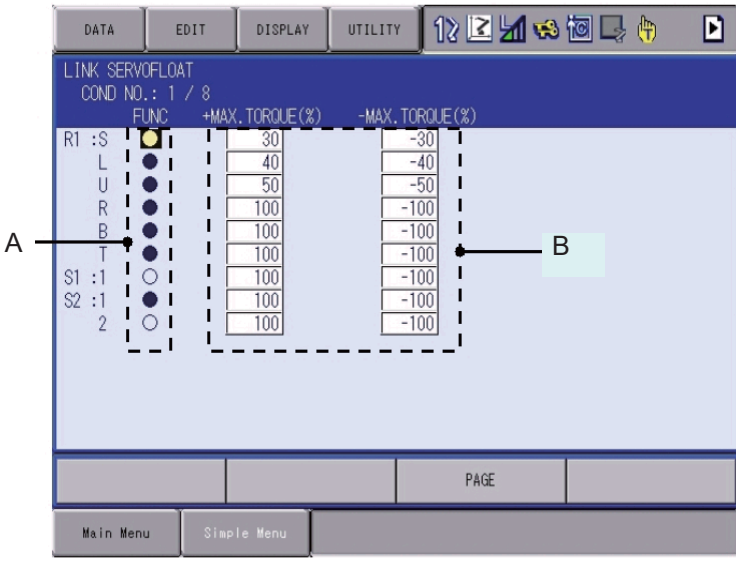
3.1.1 Procedures

1. Select {ROBOT} from the main menu.
2. Select {LINK SERVOFLOAT}
   – The link servofloat window appears.



3. Select an item to be set, and set the value.
   (1) The display enters the number input status.
   (2) Enter a value for “MAX. TORQUE” by pressing the numeric keys.
4. Press [ENTER].

3.1.2 Link Servofloat Condition Setting



A. FUNC

Enables/Disables the link servofloat function for each axis.

"●" indicates that the function is enabled; "○" indicates that the function is disabled.

Pressing [SELECT] switches between "●" and "○".

B. (+) MAX. TORQUE / (-) MAX. TORQUE

Limits the amount of torque generated for keeping control of the position of each axis.

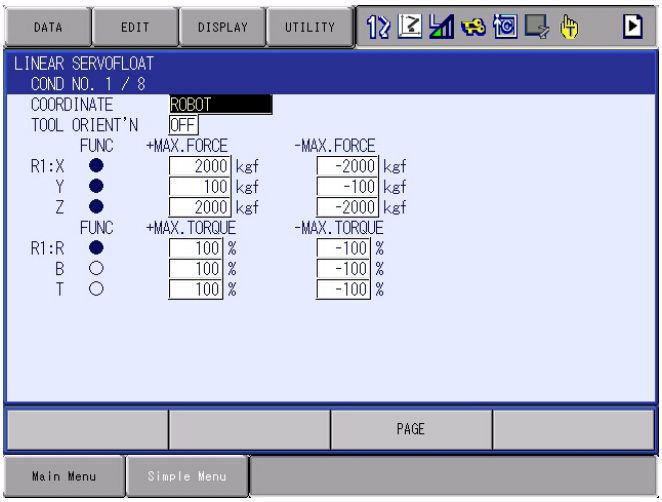
Set the value for each axis using the ratios (%) of the positive and negative sides of the motor to the motor rated torque.

The smaller the value is set, the more easily an axis can be moved by an external force.

3.2 Linear Servofloat Condition File

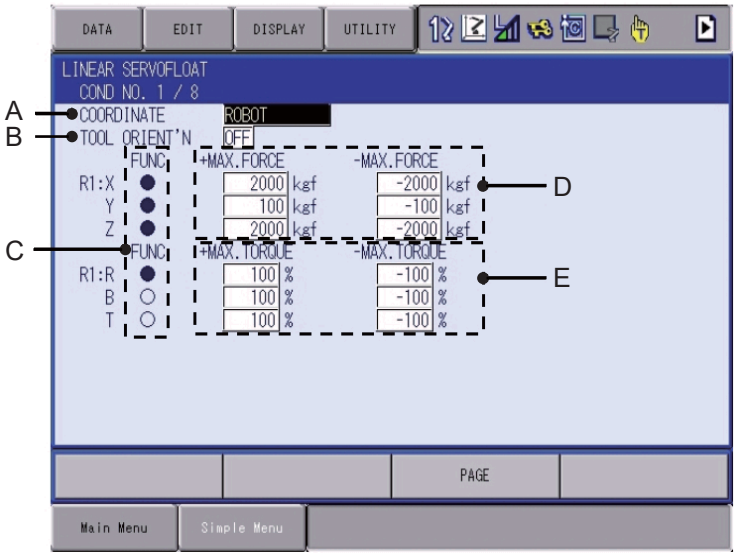
3.2.1 Procedures

1. Select {ROBOT} from the main menu.
2. Select {LNR. SERVOFLOAT}
   – The linear servofloat window appears.



3. Select an item to be set, and set the value.
   (1) The display enters the number input status.
   (2) Enter a value for “MAX. TORQUE” by pressing the numeric keys. Then, press [ENTER].
4. Press [ENTER]

3.2.2 Linear Servofloat Condition Setting



A. COORDINATE

Sets a coordinate system for the linear servofloat function.

Select the ROBOT coordinates, BASE coordinates, USER coordinates or TOOL coordinates.

The selection dialog box appears by pressing [SELECT]. Select a coordinate system to be set.

B. TOOL ORIENT'N

Enables or disables the tool orientation control for linear servofloat operations.

When the tool orientation control is enabled:
Keeps the tool tip in the same orientation during the linear servofloat operation as that at the start of the operation.

When the tool orientation control is disabled:
Does not keep the tool tip in the same orientation during the linear servofloat operation as that at the start of the operation.

“ON” indicates that the tool orientation control is enabled. “OFF” indicates that the control is disabled.

Press [SELECT] to switch between “ON” and “OFF”.

C. FUNC

Enables or disables the linear servofloat function for each axis.

"●" indicates that the function is enabled; "○" indicates that the function is disabled.

Press [SELECT] to switch between "●" and "○".

D. (+) MAX. FORCE / (-) MAX. FORCE

Limits the amount of force generated for keeping control of the position of each axis.

When the linear servofloat is enabled, a force bigger than this set value is not generated.

The smaller the value is set, the more easily a set coordinate axis can be moved by an external force.

However, if the set value is less than the manipulator friction, the manipulator may not move.

E. (+) MAX. TORQUE / (-) MAX. TORQUE

Limits the amount of torque generated for keeping control of the position of R-, B-, and T-axes.

Set the value for each axis using the ratios (%) of the positive and negative sides of the motor to the motor rated torque.

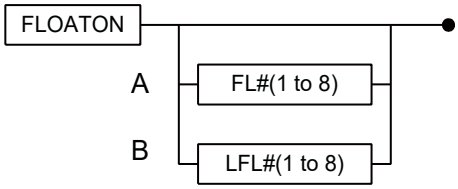
The smaller the value is set, the more easily an axis can be moved by an external force.

4 Instructions for the Servofloat Function

4.1 FLOATON Instruction

FLOATON is an instruction to enable and start the link servofloat function or the linear servofloat function.

Additional items for the FLOATON instruction are as follows:



A. FL# ( ) (Setting range: 1 to 8)

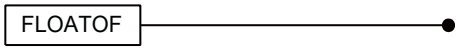
Sets the link servofloat condition file number.

B. LFL# ( ) (Setting range: 1 to 8)

Sets the linear servofloat condition file number.

4.2 FLOATOF Instruction

FLOATOF is an instruction to disable and end the link servofloat function or the linear servofloat function. The servofloat function can be also disabled by performing servo OFF.



4.3 SPDL Tag

The SPDL tag is a stop confirmation tag with which a motion is completed when the number of speed feedback of all the axes becomes less than a constant value.

Use an SPDL tag to confirm the end of a motion caused by an external force. Add it to the move instruction immediately before the FLOATOF instruction.

If the move instruction is completed with a SPDL tag added, the next instruction will not be performed as long as the manipulator is being moved by an external force.

Only “0” can be set for the SPDL tag.

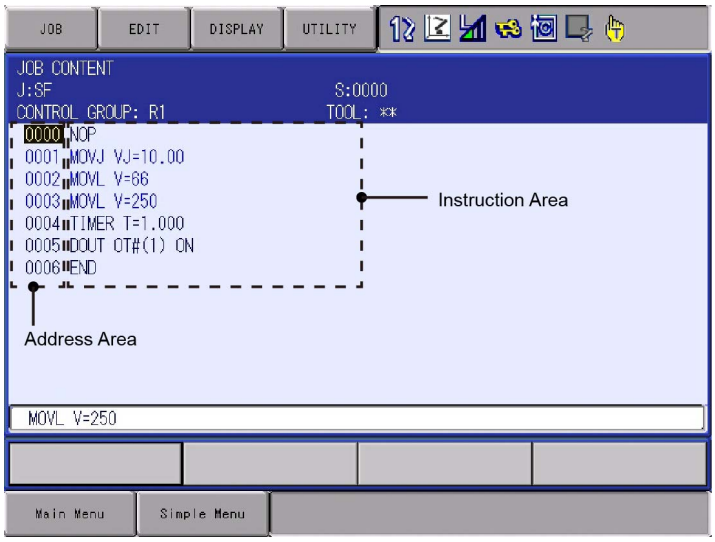
An example of an SPDL tag added to a move instruction is shown as follows:

```
MOVJ  VJ=50.00  SPDL=0
```

5 Registration of Instruction

When in the job content display in teach mode, register an instruction with the cursor in the address area.

1. Select {JOB} from the main menu.
2. Select {JOB CONTENT}
   – The job content window appears.



3. Move the cursor to the address area.

5.1 FLOATON Instruction

1. Move the cursor to the line where “FLOATON” is to be registered.
2. Press [INFORM LIST]
   – The instruction dialog box appears.

3. Select “FLOATON”
   – The “FLOATON” instruction appears in the input buffer line.

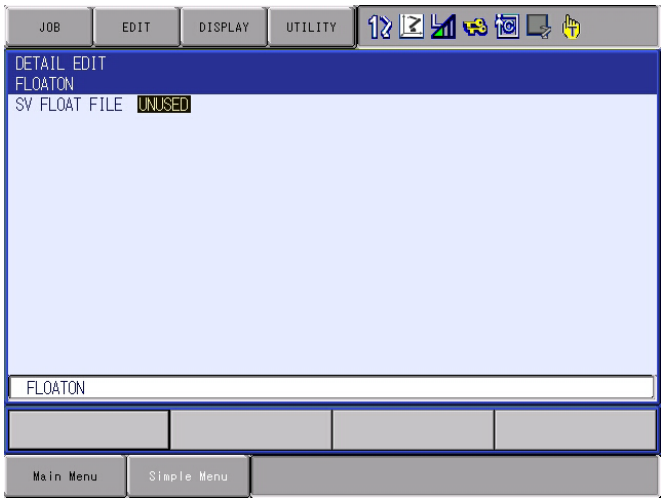
4. Change any additional items and numerical values.

– < To register without change>
To register the instruction displayed in the input buffer line without any changes, go to step 5.

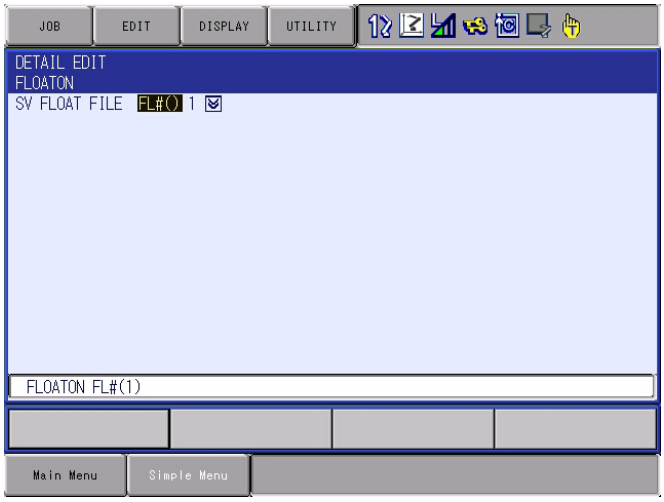
– < To edit the additional items>

(1) To add an additional item, with the cursor on the instruction in the input buffer line, press [SELECT] to call the detail edit display.

(2) With the cursor on “UNUSED” for the “SV FLOAT FILE”, press [SELECT].



– The selection dialog box appears with the choices “FL# ( )” or “LFL# ( )”. Select one.



(3) After adding an additional item, press [ENTER].

– The detail edit display closes, and the job content display reappears.

5. Press [INSERT] and [ENTER]

– The instruction displayed in the input buffer line is registered.

5.2 FLOATOF Instruction

1. Move the cursor to the line where “FLOATOF” is to be registered.
2. Press [INFORM LIST]
   – The instruction dialog box appears.

3. Select “FLOATOF”
   – The “FLOATOF” instruction appears in the input buffer line.

4. Press [INSERT] and [ENTER]
   – The instruction displayed in the input buffer line is registered.

5.3 Addition of SPDL Tag to a Move Instruction

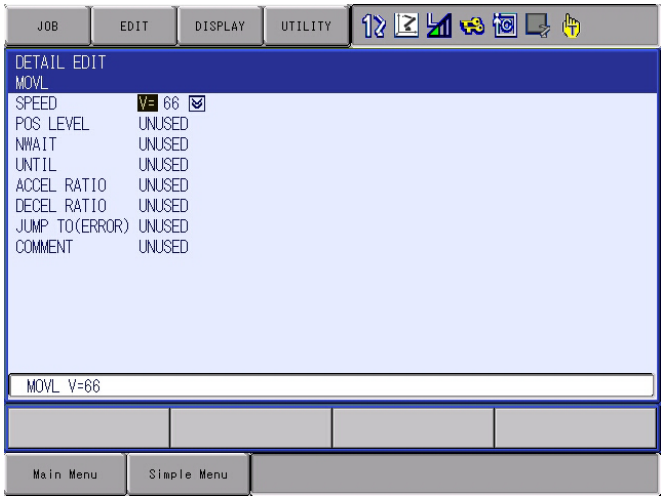
When in the job content display in teach mode, add an SPDL tag to a move instruction, with the cursor in the instruction area.

1. Select the line of the move instruction where the SPDL tag is to be added.

   – The move instruction appears in the input buffer line.

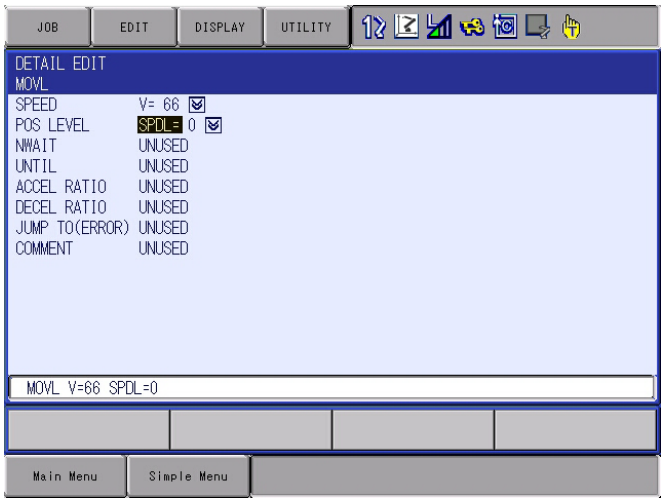
2. Press [SELECT]

   – The detail edit window appears.



   (1) With the cursor on “UNUSED” for the “POS LEVEL”, press [SELECT].

   – The selection dialog box appears with the choice, “SPDL=”.



   (2) After adding “SPDL=” to the move instruction, press [ENTER].

   – The detail edit window closes, and the job content window reappears.

3. Press [INSERT] and [ENTER]

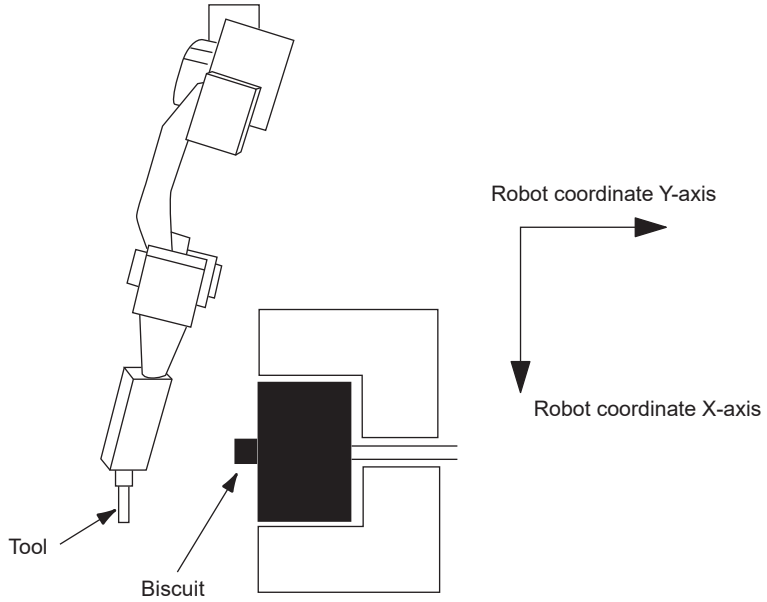
   – The instruction displayed in the input buffer line is registered.

6 Application Example of the Servofloat Function

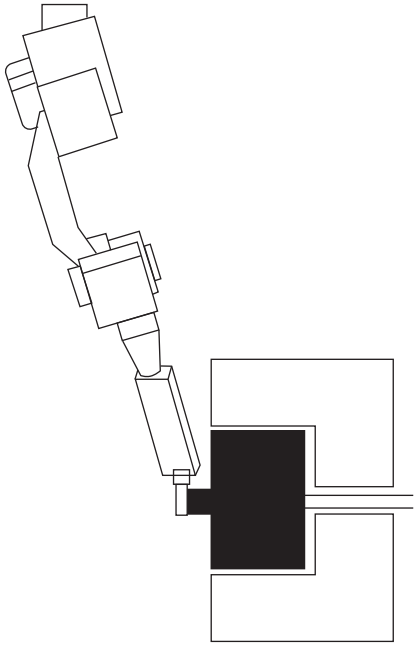
When removing a workpiece from a die-casting machine, a large external force is applied to the manipulator during extraction after grasping it. In such a case, the following example is applicable using the servofloat function.

6.1 Movement Example

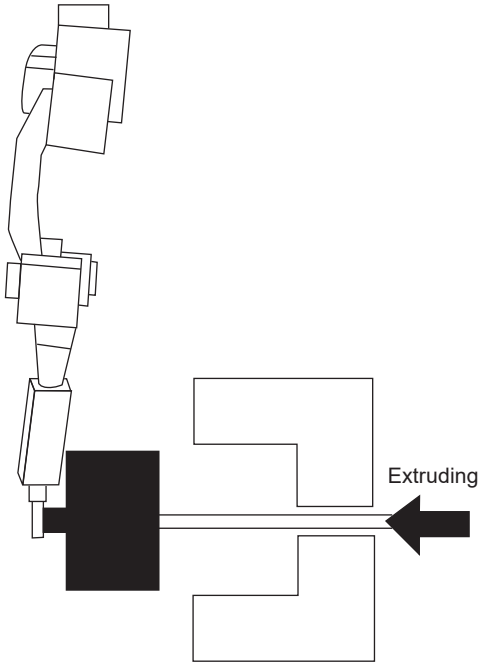
1. Moves to the waiting point.



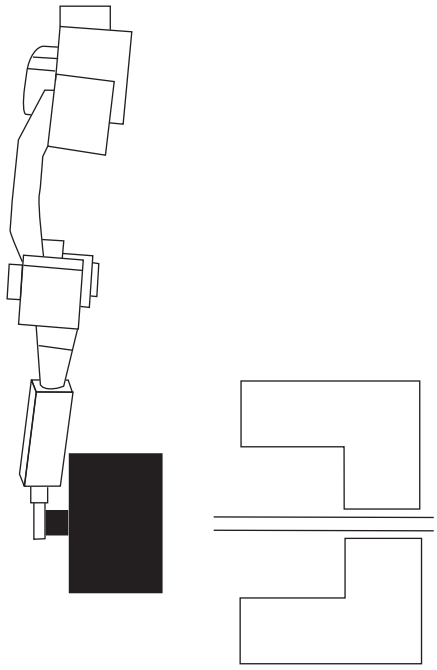
2. Moves to the grasping starting point and performs the TIMER instruction, and then starts the servofloat function.
3. When function enabled, the manipulator grasps a “biscuit”.



4. After having grasped a workpiece, the “start extraction instruction” signal is output, and the manipulator performs an extracting motion.
5. The “start extraction instruction” signal starts the extraction operation of the die-casting machine.
   – Thereby, the manipulator performs an extracting motion.
6. The die-casting machine outputs the “end extraction instruction” signal when the manipulator reaches the completion position.
   – The manipulator ends the servofloat function.



7. The manipulator returns to its normal motion and performs an extraction motion with the workpiece.
8. Then, it returns to the home position.



6.2 Job Example

No.	Explanation	Instruction
1[1)]	Moves to the waiting point.	MOVJ VJ=50.0
2	Moves to the workpiece grasping point.	MOVL V=300.0
	Stops and waits for 0.5 seconds.	TIMER T=0.500
	Starts the linear servofloat function.	FLOATON LFL#(1)
3	Turns ON the “grasping instruction” signal. (general output No. 10)	DOUT OT#(10)=1
	Waits for the “grasping answer” signal. (general input No. 10)	WAIT IN#(10)=1
4	Turns ON the “start extraction instruction” signal. (general output No. 11).	DOUT OT#(11)=1
5	Performs the extraction motion. Ends the motion when the “end extraction instruction” signal (general input No. 11) is input during the motion.	MOVL V=50.0 UNTIL IN#(11)
	Stops and waits for 0.5 seconds.	TIMER T=0.500
6	Completes the servofloat function.	FLOATOF
7	Starts the extracting.	MOVL V=300.0
	Returns to the waiting point.	MOVJ VJ=50.0

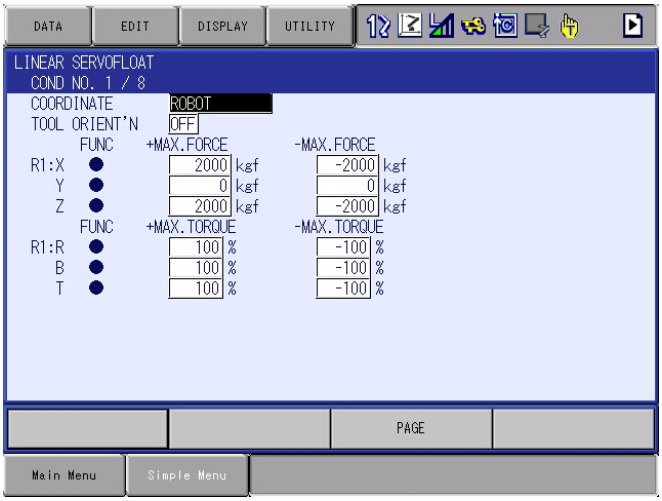
1 The numbers represent the procedure numbers in *chapter 6.1 “Movement Example”* .

6.3 A Setting Example in the Linear Servofloat Condition File

For the example described in *chapter 6.1 “Movement Example”* .

In the example, the extracting direction corresponds to that of the Y-axis of the robot coordinates.

If the direction does not correspond to the robot coordinates, define the extracting direction with the user coordinates and set “COORDINATE” to “USER#01”.

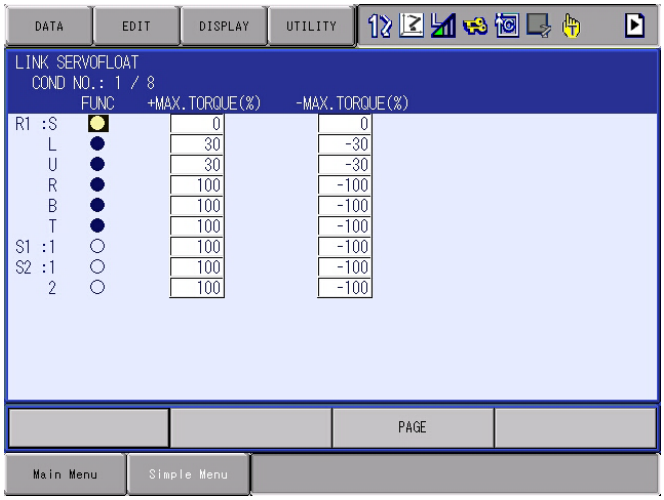


6.4 A Setting Example for the Link Servofloat Condition File

For the example described in *chapter 6.1 "Movement Example"* , specify the link servofloat condition file by specifying LFL# as the FLOATON instruction file. In this case, a setting example is as follows.

Since the S-axis moves according to the force applied by the extruding cylinder of the die-casting machine, set the "MAX. TORQUEs" of the S-axis to "0". Set the "MAX. TORQUEs" of the L- and U-axes to "30" to prevent the arms from dropping or rising up due to the load variation. To orientate the tool posture, set the "MAX. TORQUEs" of the R-, B-, and T-axes to "100".

Using the above settings as a guide, adjust the ratio of each set value according to the circumstances.



YRC1000micro OPTIONS
INSTRUCTIONS
FOR SERVOFLOAT FUNCTION

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